Facial Recognition System

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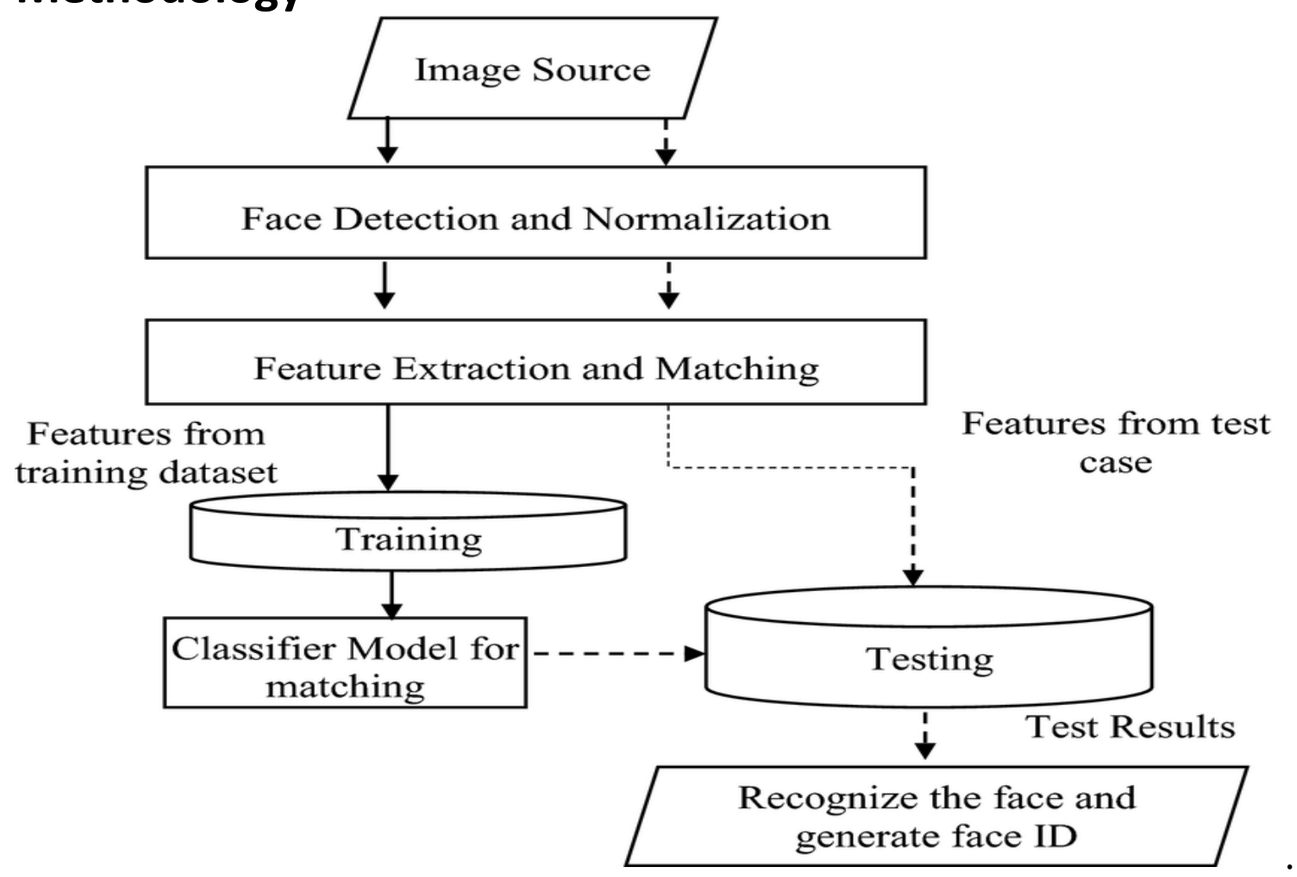
Introduction

- Built with Deep Learning – Implemented using Siamese Networks for facial recognition.
- Convolutional Neural Network (CNN) – Extracts facial embeddings for comparison.
- Real-Time Face Verification – Uses OpenCV to capture and verify faces instantly.
- Image Preprocessing – Rescales, normalizes, and converts images for model compatibility.
- Loss Function – Trained with Binary Cross-Entropy for better accuracy.
- Optimizer – Uses Adam optimizer to minimize loss and improve model efficiency.
- Evaluation Metrics – Assessed using Precision, Recall, and Accuracy.
- Scalability – Can be deployed for authentication systems, attendance tracking, and security.

Materials and methods

- Dataset – Custom dataset collected for training and verification (No external dataset used).
- Model Architecture – Siamese Neural Network (SNN) with Convolutional Neural Networks (CNNs).
- Image Processing – Utilizes OpenCV for real-time image capture and preprocessing.
- Backend – Built using TensorFlow and Keras for deep learning model implementation.
- Frontend – Uses OpenCV and Python for real-time face verification.
- Hardware – Requires a system with 4GB+ RAM, modern CPU, and a webcam for real-time detection.
- Software –TensorFlow, OpenCV, and Matplotlib for visualization.

Methodology



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Results

[1] Accurate Face Recognition – Successfully built using deep learning models for precise and reliable facial identification.

[2] Real-Time Detection – Implements OpenCV and CNNs for fast and efficient face detection in real-time.

[3] Secure Authentication – Enhances security by verifying individuals using biometric facial features.

[4] Optimized Training – Trained on a diverse dataset, improving accuracy under various lighting and facial conditions.

[5] Scalable & Adaptable – Can be extended for applications like attendance systems, security access, and smart surveillance.

Conclusions

- The system provides an efficient and secure facial recognition solution using deep learning.
- Enhances security and authentication by eliminating manual identity verification.
- Reduces errors in identification and improves accuracy through AI-driven recognition.
- Can be integrated into various applications such as surveillance, attendance tracking, and access control.
- Scalable and can be extended with anti-spoofing techniques and multi-factor authentication.

Literature cited

[1] TensorFlow Documentation
[2] Keras Functional API Guide
[3] OpenCV Official Documentation
[4] Siamese Networks for Face Recognition
[5] Deep Learning with Python – Nicholas Rennote

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